

# MD. REPON ISLAM

Assistant Professor, Department of Computer Science & Engineering  
Khulna University of Engineering & Technology (KUET), Khulna-9203, Bangladesh

repon@cse.kuet.ac.bd · +880 1521-427885 · Room CSE-210, Academic Building Block-B

Web: cse-repon.github.io · Google Scholar: scholar.google.com/citations?user=vZMGnsEAAAAJ · ORCID: 0000-0003-0416-2990 · GitHub: github.com/cse-repon

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## RESEARCH INTERESTS

Efficient Multimodal AI · Edge AI · Vision–Language Models · Computer Vision & OCR · Assistive AI · Resource-Constrained AI Systems

I study how multimodal foundation models, computer vision, and OCR can be adapted for efficient, reliable deployment on resource-constrained edge platforms, with applications in assistive technology and large-scale real-world intelligent systems. I intend to pursue doctoral research on efficient, trustworthy, and resource-aware multimodal AI (prospective PhD applicant, Spring/Fall 2027).

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## ACADEMIC APPOINTMENTS

**Assistant Professor, Dept. of CSE, KUET** Jan 2026 – present

Undergraduate teaching in artificial intelligence and embedded systems; supervision of undergraduate research and theses; industry–academia collaborative projects in AI, edge computing, and IoT.

**Lecturer, Dept. of CSE, KUET** May 2022 – Dec 2025

Part-time from May 2022; full-time from December 2022.

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## EDUCATION

**M.Sc. Engg. in Computer Science & Engineering, KUET** 2022 – 2025

CGPA 4.00 / 4.00 · Thesis: “Adapting Large Multimodal Models with Enhanced Optical Character Recognition for Vision Assistants” · Supervisor: Prof. Dr. Muhammad Sheikh Sadi

**B.Sc. Engg. in Computer Science & Engineering, KUET** 2017 – 2022

CGPA 3.93 / 4.00 · 2nd position among 120 students · Dean's Award

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## RESEARCH EXPERIENCE

**Wearable Multimodal Vision Assistant (M.Sc. research — lead researcher)** 2023 – 2025

Designed and built a hybrid edge–server vision assistant for visually impaired users: on-device OCR/ASR/TTS on a wearable capture device, a local LLM on an edge node for responsive interaction, and an 8B-parameter open-source multimodal model on a consumer-grade GPU server for complex visual reasoning. Evaluated on assistive visual question answering. Basis of the IEEE RAAICON 2024 Best Paper and two first-author publications.

**Efficient Bengali OCR & Document Understanding (supervisor)** 2025 – present

Layout-aware recognition of Bengali printed text using vision-language-model-assisted annotation, dataset construction, and domain-specific PP-OCR fine-tuning for a low-resource language. Accepted at IEEE COMPAS 2026 (student first author).

**Assistive Mobility & Sensing (researcher)** 2019 – 2022

IoT-enabled smart wheelchairs controlled by finger gesture, hand gesture, and voice; CNN-based hand-gesture human–computer interaction; deep-learning fall-detection systems. Published in Sensors, IEEE Access, SN Computer Science, and Springer proceedings.

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## LARGE-SCALE AI & EDGE SYSTEMS EXPERIENCE

Technical contributor within collaborative industry and research teams; entries are described at a high level, omitting client-sensitive details.

### **Distributed Edge AI for Mass-Gathering Operations**

2023 – 2025

Computer-vision systems in continuous production across the operational areas of one of the world's largest annual mass gatherings, over three consecutive seasons: dataset development, model training and validation, hardware-aware optimization and NPU model conversion, fleet deployment, offline caching with server synchronization, and OS/firmware optimization.

### **Large-Scale OCR & Multimodal Document Intelligence**

2023 – present

High-throughput digitization and understanding of millions of paper documents for public-sector and enterprise workflows: OCR and LLM inference on high-performance on-premise GPU clusters, vector retrieval, and multi-agentic RAG.

### **Industrial IoT & Energy-Sector Intelligence Platforms**

2023 – present

Nationwide monitoring platforms combining edge computing, industrial meter communication, centralized data acquisition, anomaly monitoring, and LLM-assisted analytics; domain-specific generative-AI assistants with grounded retrieval over organizational documents and databases.

## **PEER-REVIEWED PUBLICATIONS**

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*14 publications; first-author name in bold. Full list with metrics: Google Scholar.*

### **Journal Articles**

1. M. S. Sadi, M. Alotaibi, **M. R. Islam**, M. S. Islam, T. Alhmiedat, Z. Bassfar, "Finger-Gesture Controlled Wheelchair with Enabling IoT," *Sensors*, 22(22):8716, 2022.
2. M. M. Islam, O. Tayan, **M. R. Islam**, M. S. Islam, S. Nooruddin, M. N. Kabir, M. R. Islam, "Deep Learning Based Systems Developed for Fall Detection: A Review," *IEEE Access*, 8:166117–166137, 2020.
3. M. M. Islam, **M. R. Islam**, M. S. Islam, "An Efficient Human Computer Interaction through Hand Gesture Using Deep Convolutional Neural Network," *SN Computer Science*, 1(4):211, 2020.
4. M. M. Islam, N. H. Neom, M. S. Imtiaz, S. Nooruddin, **M. R. Islam**, M. R. Islam, "A Review on Fall Detection Systems Using Data from Smartphone Sensors," *Ingénierie des Systèmes d'Information*, 24(6):569–576, 2019.

### **Conference Papers & Proceedings**

5. A. N. E. Safa, **M. R. Islam**, M. N. I. Sakib, "A Layout-Aware Bengali Printed Text OCR Framework with Vision-Language Model-Assisted Annotation and PP-OCrv6 Fine-Tuning," *IEEE COMPAS*, 2026. (Accepted)
6. M. A. Molla, A. Kairy, **M. R. Islam**, "CycleX: IoT-Integrated Smart Bicycle Sharing System for Campus Environments," *IEEE EICT*, 2025.
7. **M. R. Islam**, M. S. Sadi, "Generative AI-assisted Standalone Wearable Reading Glasses for the Visually Impaired," *IEEE RAAICON*, pp. 173–176, 2024. **Best Paper Award.**
8. **M. R. Islam**, M. S. Sadi, "Adapting Open-source Multimodal Large Language Models for Enhanced Vision Assistants," *IEEE ICCIT*, pp. 3560–3565, 2024.
9. H. Neha, M. S. Sadi, **M. R. Islam**, S. Sakib, "Automated Object Identification for the Visually Impaired," *IEEE ICCIT*, pp. 3584–3589, 2024.
10. S. Sakib, M. S. Sadi, M. S. Arefin, **M. R. Islam**, H. Neha, "An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading," *IEEE ICECE*, pp. 767–772, 2024.
11. G. T. Roy, A. I. Atik, M. R. Rabby, **M. R. Islam**, "Enhancing Fire Safety: Encoder-Decoder Based Detection and IoT-Driven Response System," *IEEE ICICT*, pp. 219–223, 2024.
12. M. A. Rahman, M. S. Islam, **M. R. Islam**, M. S. Hossain, S. A. F. Al-Hazzaa, "An IT/OT Converged Cyber Range Framework Supporting Ensemble of Large Multimodal Models-Human Teaming and RAG

Capability for Bigdata Information Retrieval — 6G Perspective,” *IEEE GLOBECOM*, pp. 3679–3684, 2024.

13. M. A. Rahman, **M. R. Islam**, M. Rahman, “iGRC — An LLM Powered Governance, Risk, and Compliance Management Virtual Assistant for Enterprise,” *Proc. 2nd Int. Conf. on Emerging Trends in Cybersecurity*, Springer Nature, 2024. (In press)

14. **M. R. Islam**, M. S. Islam, M. S. Sadi, “Towards Developing a Real-Time Hand Gesture Controlled Wheelchair,” *Springer TCCE*, AISC vol. 1309, pp. 355–366, 2021.

AWARDS & HONORS

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- **Best Paper Award**, IEEE RAAICON 2024 — “Generative AI-assisted Standalone Wearable Reading Glasses for the Visually Impaired”.
- **Perfect CGPA 4.00/4.00**, M.Sc. Engg. in CSE, KUET, 2025.
- **Dean's Award** and 2nd position among 120 students, B.Sc. Engg. in CSE, KUET, 2022.

TEACHING & SUPERVISION

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- Undergraduate courses in artificial intelligence and embedded systems, Dept. of CSE, KUET.
- Supervision of undergraduate theses and projects, including published student research (IEEE EICT 2025; IEEE COMPAS 2026, student first author).

ACADEMIC SERVICE

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- Peer reviewer, *PLOS ONE* — three completed reviews (2024), verified via ORCID / Web of Science.

OPEN-SOURCE RESEARCH SOFTWARE

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- **RKNPU driver update for Orange Pi 5B** — most-starred repository; enables current NPU inference on affordable Rockchip hardware ([github.com/cse-repon/orangepi-5b-rknpu-0.9.8-update](https://github.com/cse-repon/orangepi-5b-rknpu-0.9.8-update)).
- YOLO RKNN inference and multi-threaded NPU pipelines for resource-constrained deployment.

TECHNICAL RESEARCH SKILLS

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|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Modeling & training | PyTorch, TensorFlow, CNNs, multimodal LLMs/VLMs, fine-tuning and adaptation, RAG                                                          |
| Vision & OCR        | OpenCV, YOLO, PaddleOCR, detection/tracking, scene-text and document pipelines                                                            |
| Edge deployment     | ONNX, RKNN/RKNPU, TensorFlow Lite, Rockchip NPUs, Raspberry Pi, NVIDIA Jetson, embedded Linux, firmware-level optimization                |
| Systems             | on-premise GPU inference (vLLM), Docker, Linux, offline-first caching and synchronization, IoT protocols (MQTT, Modbus RTU), backend APIs |
| Languages           | Python, C/C++, Java                                                                                                                       |

REFERENCES

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|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prof. Dr. Muhammad Sheikh Sadi</b><br>Professor, Dept. of CSE, KUET<br>(M.Sc. thesis supervisor)<br>sadi@cse.kuet.ac.bd | <b>Prof. Dr. Mohammed Abdur Rahman</b><br>Full Professor & Founding Head, Cyber Security<br>and Forensic Computing, Univ. of Prince Mugrin<br>(research collaborator) |
| <b>Prof. Dr. Al-Mahmud</b><br>Professor & Head, Dept. of CSE, KUET<br>mahmud@cse.kuet.ac.bd                                |                                                                                                                                                                       |